

ASSISTED REPRODUCTIVE TECHNOLOGIES

ART

GOALS

Understand the purpose and processes of the three different types of Assisted Reproductive Technologies,

- * *In-Vitro* Fertilisation,
- * Intra-Uterine Insemination
- and * Embryo Transfer

WARM UP



LET'S REVIEW OVULATION AND FERTILISATION.

After watching the video....

1. What is the purpose of the FSH hormone?

- (a) Prepares the uterus for a fertilised egg
- (b) Ripens a few follicles in the ovaries
- (c) To release the egg from the ovary
- (d) Start menstruation.



2. How many follicles become fully mature in a normal monthly cycle?

- (a) One
- (b) Two
- (c) Three
- (d) More than three.

3. Which hormone is necessary to then maintain a pregnancy?

- (a) Oestrogen
- (b) Progesterone
- (c) FSH
- (d) LH



Important.
Please learn!



Definition

Infertility is the inability to conceive or sustain a pregnancy.

15% of couples will have infertility problems.

Read and take notes



CAUSES OF INFERTILITY.

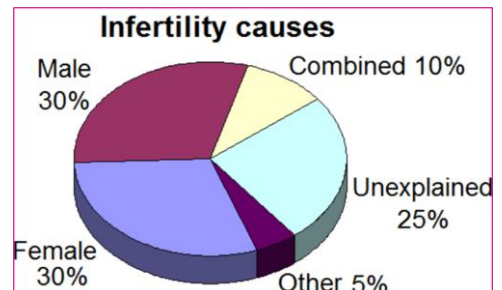
Main factors affecting infertility for a woman...

- **Ovulatory** (e.g. when a woman does not release an egg every month; poor egg quality)
- **Tubal** - tubal blockage (e.g. STI's / Pelvic Endometriosis)
- **Age**

Main factors affecting infertility for a male...

Semen:

- volume concentration,
- motility,
- morphology



IMPACT OF AGE

For women fertility starts to slowly decline when they are in their early thirties. Around age 35 the decline speeds up and **by age 40 the chance of conceiving has fallen by half.** The biological reason for this is...

- As women age, **the total number of eggs in their ovaries and their quality decline.**
- The average age of menopause is about **51 years** (and this has not changed over the centuries) and this marks the **absolute end of fertility**
- It is **rare for a woman to conceive in the 10 years leading up to menopause.**
- At age **40 about 75% of eggs are chromosomally abnormal** which **reduces the chance of pregnancy** and **increases the risk of miscarriage**

HOW TO TREAT INFERTILITY.

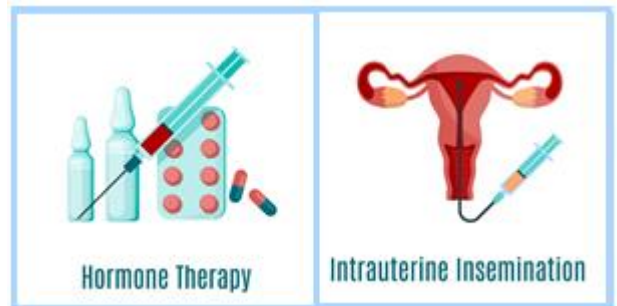
- Surgery (*not discussed*)
- Ovulation induction
- Intra - uterine insemination
- In-vitro fertilisation.



OVULATION STIMULATION AND INDUCTION

The goal is to stimulate egg production, and monitor in order to maximise success of a single pregnancy.

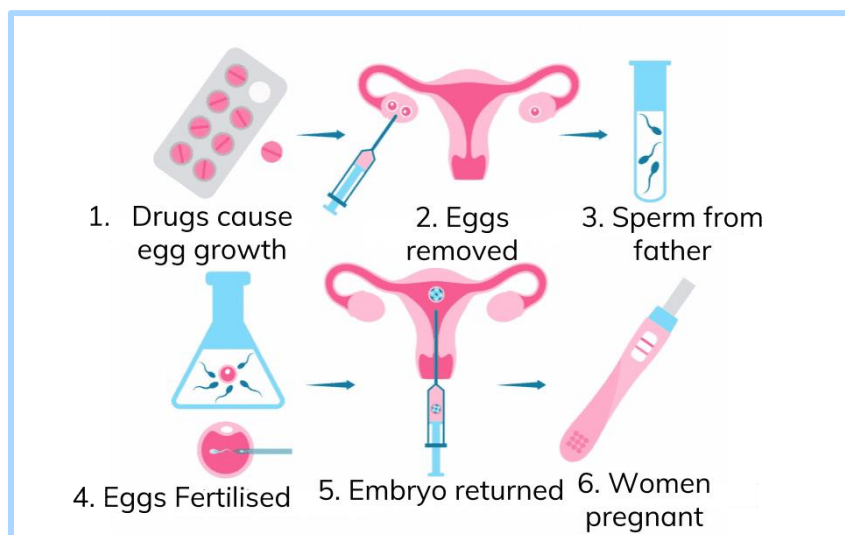
- **Injections of FSH given to increase number of eggs produced.**
- **Ovulation monitored.**
- **Normal fertilisation occurs or**
- **Intra-uterine insemination is used at time of ovulation.**



IN-VITRO FERTILISATION

Goal is to find the single best embryo in order to maximise success of a single pregnancy.

- **Egg production is stimulated.**
- **Eggs are collected and removed.**
- **External fertilisation occurs.**
- **Embryo is screened.**
- **Embryo is transferred back into uterus.**





ACTIVITY 1

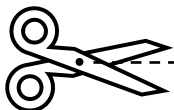
Cut out the four steps, and place in the empty boxes.
Wait until your teacher checks your work before pasting.

On average, only one follicle becomes fully mature each month.

The follicle ruptures and releases the egg which is picked up by finger like projections of the fallopian tubes.

A man takes 2-3 months to make the sperm.

hCG is secreted from the developing placenta, rescuing the follicle from degenerating. This then produces progesterone which is required to maintain the pregnancy.



It takes about **3 days** for the egg to reach the uterus, and **another 3 days** to implant into the inner lining.

The follicle makes oestrogen, which when in high enough levels, the brain releases LH, which triggers ovulation. .

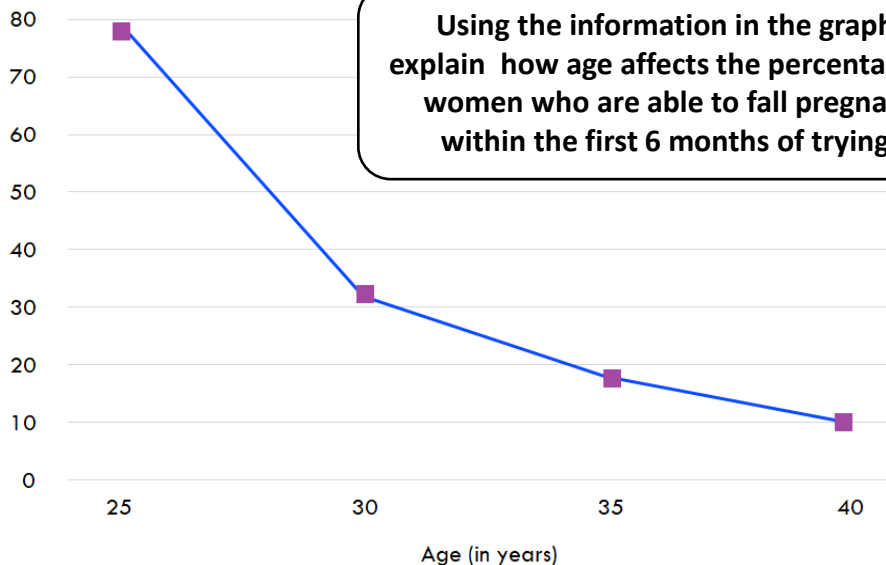
The egg must be fertilised by sperm within **24 hours** of ovulation.

Each ejaculation releases more than **100 million sperm**, but only **100** make it to the egg, and only one is needed to fertilise the egg.

ACTIVITY 2



% Pregnant
in 6 months



Using the information in the graph, explain how age affects the percentage of women who are able to fall pregnant within the first 6 months of trying.



ACTIVITY 3

In a sentence or two, compare these numbers using your own words

Risk of Chromosomal Abnormalities at Live Birth, According to Maternal Age.

Maternal Age at Delivery (year)	Risk of Chromosomal Abnormality
20	1/256
25	1/476
30	1/385
35	1/192
40	1/66
45	1/21

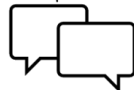


ACTIVITY 4

Infertility problems happen on an average to 1 in 8 couples.

How would the following affect the ability to conceive a child, and which ART would you recommend?

Talk to your
elbow partner



Write in
notebook



1. A 39 year old female is trying to conceive and cannot fall pregnant.

3. A 28 year old female has blocked fallopian tubes.

2. A male has a low sperm count.

